RESEARCH STATEMENT

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My research is driven by a central question: **how can we rigorously evaluate, interpret, and secure increasingly capable AI systems before they fail in high-stakes settings?** Across my work in empirical machine learning, large-model evaluation, and mechanistic interpretability, I have consistently focused on understanding model behavior under pressure: distribution shift, adversarial manipulation, weak grounding, and representation-level failure. I am especially interested in research at the intersection of **AI Safety, security, and interpretability,** where empirical analysis can expose hidden vulnerabilities and inform more robust, aligned systems.

Overview

My early research experiences built my foundation in designing reliable machine learning pipelines for noisy real-world data. In my work on remote physiological sensing, I helped develop a real-time heart rate estimation pipeline from facial RGB video using remote photoplethysmography and frequency-domain signal analysis [1]. This project taught me the importance of reproducibility, careful validation, and system-level thinking when extracting meaningful signals from imperfect data. It also shaped my broader research instinct: I became less interested in benchmark performance alone and more in interested in **why systems work, when they fail, and how to characterize those failure systemactically.**

I further developed this empirical mindset during my research at Colorado State University. At the Conrad AI Lab, I worked on noisy, high-dimensional agriculture sensor data and used unsupervised and statistical techniques such as PCA and clustering to uncover peak-detection related patterns. This experience strengthened my ability to independently scope research questions, analyze complex datasets, and translate findings into clear scientific narratives. I found myself asking not only whether a model separated classes accurately, but also what signal it was relying on and how brittle that signal might be under changing conditions.

That question became central in my later work at the Dao Hyper AI Lab (now ReSEIL Lab), where I led an end-to-end empirical evaluation pipeline for early sugar beet disease detection using hyperspectral imaging. I designed and handled preprocessing of model development, stress-tested edge cases, and contributed to the final write-up of a peer-reviewed publication. Across 122 inoculated plants, our framework achieved a ~92% accuracy in disease identification and ~96% accuracy in quantification. What mattered most to me in that project, however, was not only the final performance, but the evaluation process itself: probing class imbalance, out-of-distribution behavior, and failure cases made clear that **robust evaluation is essential to trustworthy machine learning.** This conviction now shapes how I approach frontier AI systems. [2]

My current work moves directly into the space of AI safety and security. As an AI Safety Researcher in the Algoverse AI Research Program, I investigate **mechanistic interpretability in cross-modal vision language systems,** focusing on how neural representations encode and transfer meaning across modalities. I am particularly interested in modal alignment failure, latent representation drift, and safety-relevant behaviors that emerge when internal representations become unstable or semantically misaligned. This work has strengthened my interest in using representation-level analysis to study model reliability, especially in settings where capable systems generalize in way that are difficult to predict from surface behavior alone.

In parallel, I have been building and LLM security evaluation system centered on retrieval-augmented generation. After initially exploring fine-tuning based approaches, I rescoped the system around RAG once it became clear that grounded retrieval better fits the structure and depth of security knowledge. It showed me that many model failures are not isolated bugs, but symptoms of deeper weaknesses instruction handling, grounding, system boundaries, and robustness under adversarial interaction.

These experiences are what draw me strongly to doctoral research in AI safety and security. I want to study **how advanced AI systems fail, how those failures can be detected through careful evaluation and interpretability methods, and how stronger defenses and control mechanisms can be designed in response.** I am particularly interested in questions such as: How can we identify safety-relevant failure modes before deployment? How do internal representations shift when models encounter adversarial or misleading context? What kinds of evaluation frameworks best capture prompt injection risk, contextual safety failures, task drift, or brittle reasoning behavior? And how can insights from interpretability and empirical stress-testing inform safer, more steerable AI systems?

Future Vision

I am especially excited by the opportunity to work with you because your research on **AI security, safety, alignment, prompt injection, evaluation, contextual integrity, representation engineering, and agentic risks aligns** closely with both my current experience and the direction I want to pursue at the PhD level. I am particularly drawn to your group's focus on understanding AI failure modes while also developing practical mitigations, defenses, and control methods. That combination of empirical diagnosis and safety-oriented intervention strongly matches how I think about research. My background in mechanistic interpretability, adversarial evaluation, RAG system behavior, and reproducible experimentation would allow me to contribute meaningfully to this research environment while continuing to grow toward deeper work on language-model security, safe agents, and alignment.

Final Remarks

Overall, I am seeking a PhD environment in which I can deepen my work on AI safety, model behavior, and robust evaluation while contributing to research with clear technical and societal importance. My experience across mechanistic interpretability, adversarial evaluation, and empirical ML system design has prepared me to take on open-ended research problems with independence and rigor. I would be excited by the opportunity to contribute to your group's work and to further develop research at the intersection of AI security, safety, and alignment.

References

[1] **P. V. P. Sai Kiran,** S. Reddy Vangala and M. Rashid, "Efficient Approach for Measuring Heart Pulse from Real Time Facial RGB Color Videos," 2021 2nd International Conference on Intelligent Engineering and Management (ICIEM), London, United Kingdom, 2021, pp. 412-416, doi: 10.1109/ICIEM51511.2021.9445283.

[2] Abdalla, A., Todd, O. E., **Pennam, S. V.**, Hoopes, E., Nguyen, N. T., Dorn, K. M., & Dao, P. D. (2025). All-in-one machine learning framework for early detection and characterization of sugar beet diseases using hyperspectral imaging. *Smart Agricultural Technology*, 101633.